

2.6 Integration of Metabolism

▼ What are the five main metabolic tissue types in the body?

- Heart
- Brain and nervous tissue
- Liver
- Adipose
- Muscle

▼ Which two molecules does muscle rely on for ATP?

- Carbohydrates and fatty acids

▼ What adaptation do muscle cells make in response to moderate exercise conditions?

- There are an **increased number of glucose transporters** on the surfaces of the muscle cells allowing more to be taken up by them for glycolysis

▼ How does metabolism adapt to the requirements of vigorous muscle contraction?

- During light contraction, the ATP requirements are met by oxidative phosphorylation
- During vigorous contraction, **oxygen becomes a limiting factor** and so ATP is made via **muscular glycogen breakdown** into pyruvate which can be used in **lactate formation** (which enables NAD⁺ production and hence glycolysis, producing ATP)

▼ Which molecules do the brain and nervous tissue rely on for ATP?

- **Glucose** mainly, in some cases it can use **ketone bodies**.

▼ Which molecules can the brain and nervous tissue NOT rely on for ATP?

- **Fatty acids**.

▼ Why is hypoglycaemia in the brain dangerous?

- It can cause **comas and faintness**

▼ Why is hyperglycaemia in the brain dangerous?

- It can cause **irreversible damage**.

▼ What role does adipose tissue play in metabolism?

- The **long term storage of triglycerides**.

- ▼ Which molecules do the heart tissue rely on for ATP?
 - **Fatty acids**
 - **Carbohydrates**
 - **Ketone bodies**
- ▼ What is a structural adaptation of heart tissue which allows it to effectively perform its function?
 - It has **lots of mitochondria** which can be used for aerobic metabolism
 - This is important as the **heart is designed solely for aerobic metabolism**, since it constantly needs energy to contract
- ▼ How does skeletal muscle tissue cope with the demands placed on it by vigorous muscle contraction?
 - Oxidative phosphorylation can no longer meet its requirements during vigorous contraction
 - Hence, **skeletal muscle uses anaerobic respiration to produce lactate from pyruvate which helps it to regenerate NAD⁺**
 - **NAD⁺ can be used in glycolysis** to continually make ATP
- ▼ What role does liver tissue play in metabolism?
 - **Carbohydrate (glycogen) storage**
 - **Gluconeogenesis**
 - **Lipoprotein production** (VLDL and HDL)
- ▼ Which tissues are involved in resting metabolic rate?
 - **Brain**
 - **Heart**
 - **Liver**
- ▼ How do the liver and pancreas work hand in hand to maintain blood glucose levels?
 - Liver is involved in absorption of glucose and its conversion to glycogen so that it can be stored compactly
 - Pancreas is involved in the release of hormones insulin and glucagon which decrease and increase blood glucose levels respectively
- ▼ How and where can excess glucose-6-phosphate be stored?
 - As **glycogen** in the **liver and muscles**.

▼ How and where can excess acetyl CoA be stored?

- As **triglycerides** in the **adipose tissue**
 - **Acetyl CoA → Fatty acids via lipogenesis**
 - **Fatty acids + glycerol = Triglycerides**

▼ What happens to acetyl CoA during fasting?

- It is converted into **ketone bodies**
- This is **so there are enough fuel sources for the brain** in the fasting state

▼ What is glycogen?

- It is a **polymer of glucose** which consists of 1-4 and 1-6 glycosidic bonds.

▼ How can nucleotides be produced from carbohydrate metabolism?

- Via the **pentose-phosphate pathway (PPP)** from **glucose-6-phosphate**

▼ How can glycogen be produced from carbohydrate metabolism?

- From **excess glucose-6-phosphate**

▼ How can ketone bodies be produced from carbohydrate metabolism?

- From **acetyl CoA** in the **fasting state**
 - In the **fasting state**, **most acetyl CoA is converted to ketone bodies instead of entering the TCA Cycle**

▼ How can amino acids be produced from carbohydrate metabolism?

- From **pyruvate** and **TCA Cycle intermediates**

▼ How can fatty acids be produced from carbohydrate metabolism?

- From **acetyl CoA** via **lipogenesis**
 - **Malonyl CoA + Acetyl CoA**

▼ How can cholesterol be produced from carbohydrate metabolism?

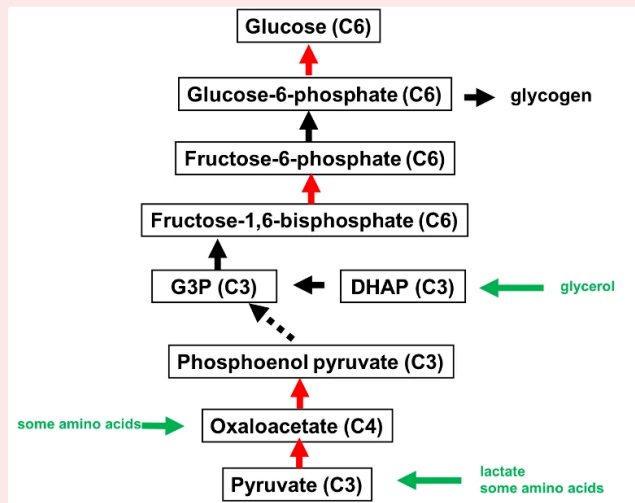
- From **acetyl CoA**
 - **Step 1: Acetyl CoA + Acetyl CoA = Acetoacetyl CoA**

▼ What is the purpose of gluconeogenesis?

- To **generate glucose for the brain** during fasting periods when it may get dangerously low
- It is an alternative to glycogen breakdown and acetyl CoA conversion to ketone bodies which can be used to prevent **hypoglycaemic comas**

▼ How can glycerol be used in gluconeogenesis?

- It is converted to **dihydroxyacetone phosphate (DHAP)**
- This is an intermediate in glycolysis so things become easy from there



▼ When exactly is lactate needed to be generated from pyruvate?

- When the rate of glycolysis exceeds the rate of the TCA Cycle + oxidative phosphorylation
- This can happen in cases where oxygen is the limiting factor (i.e during exercise) because oxidative phosphorylation requires O₂ to function.

▼ What is the Cori cycle?

- The process which involves the “re-conversion” of lactate back to pyruvate using **lactate dehydrogenase (LDH) in the liver**
 - This pyruvate can be used to **regenerate glucose** in **gluconeogenesis**
- The **regenerated glucose** is then metabolised again anaerobically to **reproduce lactate** and **regenerate NAD⁺**
- Overall, lactate is converted to glucose which is metabolised again to produce pyruvate and then lactate again. Then it all starts again.

▼ What is aim of the Cori cycle?

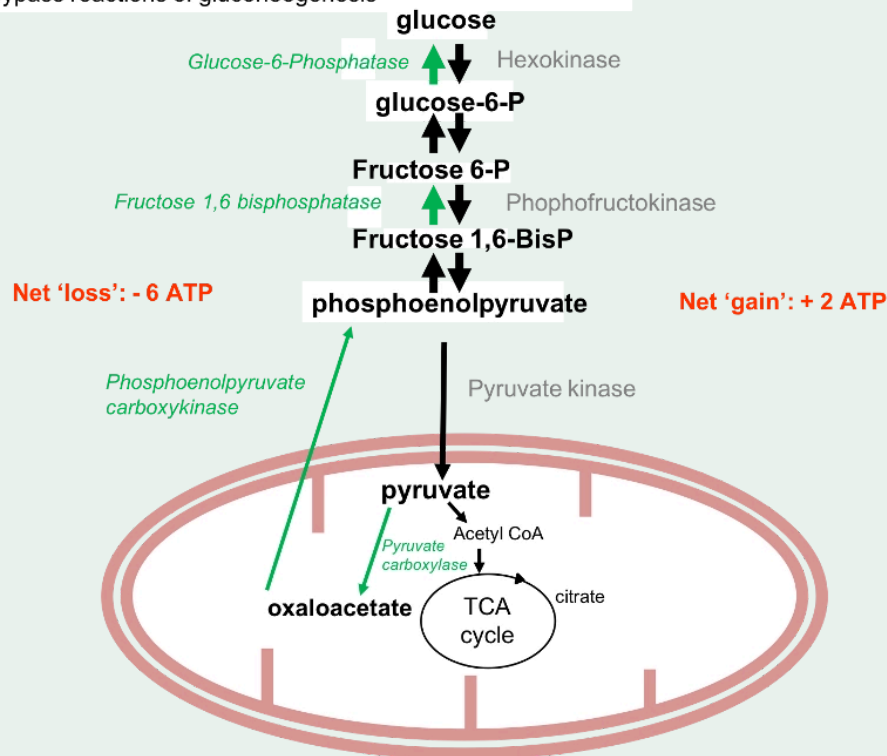
- The purpose of the Cori cycle is to **provide energy to the muscles during exercise while removing lactate and preventing lactic acidosis.**

▼ How is lactate used in gluconeogenesis?

- **Lactate** is used to **regenerate pyruvate** which can enter **gluconeogenesis**

- ▼ How are amino acids used in gluconeogenesis?
 - **Some amino acids** are used to **regenerate oxaloacetate** which can **enter gluconeogenesis**
 - **Some amino acids** are used to **regenerate pyruvate** which can **enter gluconeogenesis**
- ▼ How is glycerol used in gluconeogenesis?
 - **Glycerol** is used to **regenerate dihydroxyacetone phosphate (DHAP)** which can enter **gluconeogenesis**
- ▼ How does gluconeogenesis become an energetically favourable reaction despite delta G being positive for glycolysis reversal?
 - **Gluconeogenesis uses energy from six phosphoanhydride bonds (6x ATP) to make the reaction energetically favourable**

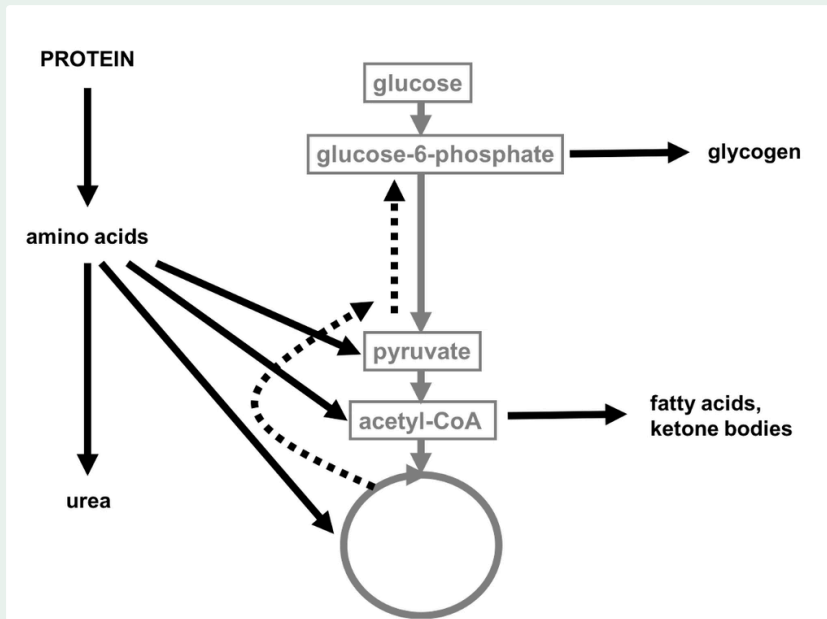
Bypass reactions of gluconeogenesis



- ▼ Which reactions need to be bypassed in order for gluconeogenesis to go ahead (because they are irreversible)?
 - Those **glycolysis** reactions **catalysed by kinases** (**hexokinase (I)**, **phosphofructokinase (III)**, **pyruvate kinase (X)**)
 - Also, the reaction in the **mitochondria** that **produces oxaloacetate from pyruvate** using **pyruvate carboxylase**

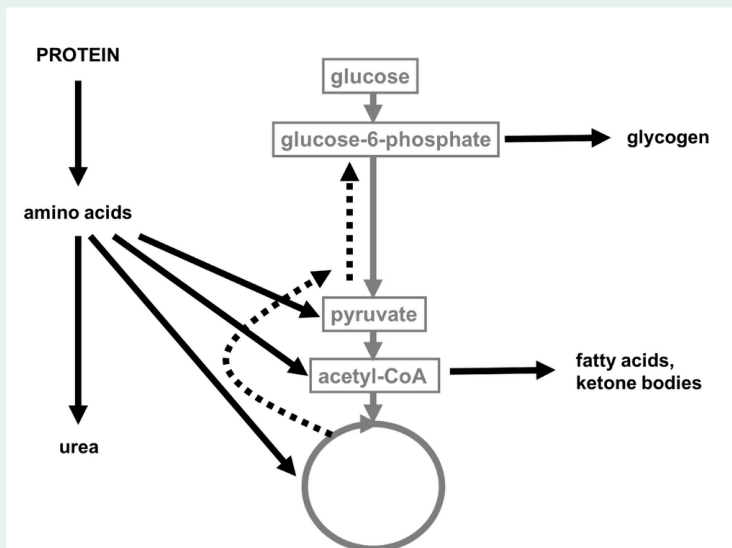
▼ What is a glucogenic amino acid?

- An amino acid that is capable of forming glucose via gluconeogenesis



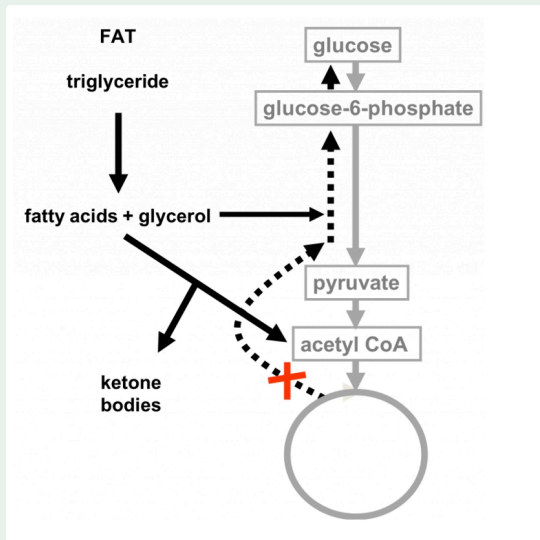
▼ What is a ketogenic amino acid?

- Amino acids that can be used to generate **fatty acids or ketone bodies**



▼ How can fats be used as fuel sources in humans?

- Triglycerides are broken down into fatty acids + glycerol
- Glycerol can then be used to generate DHAP and hence glucose via gluconeogenesis
- **FATTY ACIDS ARE NOT ABLE TO GENERATE GLUCOSE** but can be used as fuel via **beta-oxidation**



▼ What are two requirements on muscle cells that allow them to contract?

- Increased **ATP**
- Increased **number of glucose transporters**

▼ What are the two ways in which adrenaline helps muscle to cope with increased ATP demands during aerobic respiration?

- **Increased gluconeogenesis** by the liver which **increases rate of glycolysis**
- **Increased fatty acid release by adipocytes** which **increases rate of lipolysis**
- **Both of these mechanisms involve increasing ATP levels**

▼ How can lactate help humans to cope with increased demands of exercise?

- Lactate production from pyruvate **regenerates NAD⁺** which can allow glycolysis to continue
- Pyruvate production from this lactate can then be used in gluconeogenesis to form more glucose
- This is **the Cori Cycle**

▼ What is the typical range for blood glucose?

- **4.0-5.5 mM/L**

- ▼ Where are the two main isoforms of hexokinases found?
 - **Muscle (I) and liver (IV)**
- ▼ What is Michaelis constant (K_m)?
 - The **concentration of substrate** which allows an **enzyme to function at half its maximal rate**
 - It can be used to compare relative enzyme activity
- ▼ Which of the two hexokinase isoforms are more active and explain how you know this?
 - HK I
 - HK I has a K_m of around 0.1 mM but HK IV has a K_m of around 0.4 mM
 - This means that HK I is more sensitive to glucose and so is more active at lower concentrations
- ▼ Where is negative feedback involved in the control of glucose-6-phosphate production from glucose?
 - **Muscle**
- ▼ Describe how negative feedback is involved in the control of glucose-6-phosphate production from glucose.
 - **Glucose-6-phosphate (G-6-P)**, the product of this reaction, **inhibits Hk I, preventing further conversion of glucose into G-6-P**
- ▼ What is glucose-6-phosphatase used for?
 - Catalysis of the conversion of **glucose-6-phosphate back to glucose** (reversal of glycolysis Step 1)
- ▼ How does insulin work?
 - It stimulates uptake and use of glucose **when glucose levels are high**
 - It also **stimulates its storage as glycogen and fat**
- ▼ How does glucagon work?
 - It is secreted when glucose levels fall and stimulates **production of glucose via gluconeogenesis**
 - It also **stimulates breakdown of glycogen and fat**
- ▼ How does adrenaline work?
 - It has strong and fast metabolic effects to **mobilise glucose and fatty acids for 'fight or flight'**
 - It increases **lipolysis and gluconeogenesis** and hence **muscular glycolysis during exercise**

▼ How do glucocorticoids work?

- They are steroid hormones used to increase the **synthesis of metabolic enzymes concerned with glucose availability**
- Essentially used to increase and maintain glucose levels

▼ In the fed state, as well as glycogen synthesis increasing in muscle and liver cells, what other metabolic processes are stimulated?

- **Glycolysis (in liver cells)** which leads to
- **Acetyl-CoA production** and
- **Lipogenesis** (from acetyl CoA)
- Overall, **anabolic pathways** are stimulated in the fed state in liver cells, to **save energy for later**

▼ Which organ is involved in the secretion of glucagon and insulin?

- **Pancreas**

▼ In the fasting state, describe what happens in terms of hormones and energy.

- Glucagon secretion increases, insulin secretion decreases leading to glycogen → glucose and gluconeogenesis
- **Fatty acids break down** as an **alternative for ATP production** to preserve glucose for the brain
- **Glycolysis** and **lipolysis** in skeletal muscle increases due to adrenaline
- **Gluconeogenesis** and **glycogen breakdown** increase

▼ In the fasting state, what role does adrenaline play?

- **Adrenaline** stimulates glycogen breakdown as well as **glycolysis and lipolysis**

▼ What further changes occur in the starving (prolonged fasting) state?

- **Adipose tissue hydrolyses triglycerides** to provide **fatty acids for metabolism (beta-oxidation)**
- **TCA Cycle intermediates reduced** to provide **substrates for gluconeogenesis**
- **Protein breakdown** provides **amino acid substrates for gluconeogenesis**
- **Ketone bodies** are **produced from fatty acids and amino acids** to temporarily **aid the brain's glucose requirements** (brain is picky so this only happens when necessary)

- ▼ Why are fatty acids broken down for ATP production instead of glucose in the fasting state?
 - To **preserve as much glucose as possible for the brain**
 - After prolonged periods of fasting, fatty acids are converted to ketone bodies to partially substitute the glucose requirements
- ▼ What is the difference between Type I and Type II diabetes?
 - Type I diabetes is a failure to secrete enough insulin
 - Type II diabetes is a failure to respond appropriately to insulin levels.
 - Both have the effect of blood glucose levels getting too high, and individuals being treated as though they are in the starving state
- ▼ What is ketoacidosis?
 - When too many ketone bodies are produced
 - This happens to diabetes mellitus sufferers
- ▼ What tissue is the major source of ketone bodies in the fasting state?
 - **Liver**
- ▼ In the fed state, insulin acts to increase the synthesis of which molecule?
 - **Proteins** via protein synthesis
- ▼ In the fed state, why does insulin stimulate an increase in the number of